10	**(a)**		$\frac{1}{(4x+1)(x+1)} = \frac{A}{4x+1} + \frac{B}{x+1}$ $1 = A(x+1) + B(4x+1)$ $x = -1 \Rightarrow 1 = -3B \Rightarrow B = -\frac{1}{3}$ $x = -\frac{1}{4} \Rightarrow 1 = \frac{3}{4}A \Rightarrow A = \frac{4}{3}$ So $\frac{1}{(4x+1)(x+1)} = \frac{4}{3(4x+1)} - \frac{1}{3(x+1)}$	**M1** **A1** **A1** **[3]**	**1.1a** **1.1** **1.1**	Method mark is implied by correct answer. Final solution needed for A1. Can be recovered in 10(b)	
10	**(b)**		$\int \frac{1}{y}\,dy = \int \frac{1}{(4x+1)(x+1)}\,dx$ $\int \frac{1}{y}\,dy = \int \left\{\frac{4}{3(4x+1)} - \frac{1}{3(x+1)}\right\} dx$ $ln\|y\| = \frac{1}{3}ln\|4x+1\| - \frac{1}{3}ln\|x+1\| + c$ When $x = 0, \quad y = 2 \Rightarrow c = ln\,2$ $ln\|y\| = ln\,2\left(\frac{\|4x+1\|}{\|x+1\|}\right)^{\frac{1}{3}}$ $y = 2\left(\frac{4x+1}{x+1}\right)^{\frac{1}{3}}$	**M1** **M1** **M1** **B1** **M1** **A1** **[6]**	**3.1a** **2.2a** **1.1** **2.2a** **1.1** **2.1**	Separation of variables - both sides seen Use of *their* **(a)**. RHS only needed Integration. One correct *x* term (ft their partial fractions) Condone missing modulus signs Finding constant (FT *their* **(a)**) For use of $aln\,m = ln\,m^a$ and $ln\,m - ln\,n = ln\frac{m}{n}$ Not dep on *c* Condone missing modulus signs throughout Answer with correct values of *A* and *B*.	Integral signs, d*y* and d*x* needed Condone no d*x*